



1. What is the value of $\frac{\sqrt{2}-3}{2+\sqrt{3}} + \frac{\sqrt{2}+3}{2-\sqrt{3}}$?

- (A) $3\sqrt{2} + 2\sqrt{3}$ (B) $2\sqrt{2} + 3\sqrt{3}$ (C) $4\sqrt{2} + 3\sqrt{3}$ (D) $3\sqrt{2} + 4\sqrt{3}$ (E) $4\sqrt{2} + 6\sqrt{3}$

2. How many ordered quadruples of integers (w, x, y, z) satisfy the equation

$$w^2 + x^2 + y^2 + z^2 = 2?$$

- (A) 6 (B) 12 (C) 18 (D) 24 (E) 36

3. Euler and Fermat estimate the value of the following sum:

$$3.141 + 2.718 + 1.080 + 1.741.$$

Fermat estimates the sum by rounding each addend to the nearest whole number, then adding the resulting estimates. On the other hand, Euler estimates the sum by rounding each addend to the nearest tenth. How many times closer is Euler to the exact answer than Fermat?

- (A) 2 (B) 4 (C) 5 (D) 6 (E) 10

4. What is the sum of the real values of x which satisfy

$$\frac{x^4 - 3x^2 + 2}{x^2 - 3x + 2} = 0?$$

- (A) -3 (B) -1 (C) 0 (D) 1 (E) 3

5. Two grandparents, two parents, a pair of twin girls and a boy decide to take a family photo. In how many ways can they arrange themselves in a line if the boy insists on standing in between either both the twin girls, both the parents, or both the grandparents?

(A) 180 (B) 240 (C) 360 (D) 720 (E) 1080

6. Dan has two equally-sized cups labeled A and B . Cup A is half-filled with tea and cup B is half-filled with milk. He pours $\frac{1}{3}$ of the contents of cup B into cup A and mixes the contents of cup A . Afterwards, he pours $\frac{2}{5}$ of the contents of cup A into cup B and mixes the contents of cup B . Given that $m\%$ of cup A is tea and $n\%$ of cup B is milk, what is m/n ?

(A) $\frac{3}{4}$ (B) $\frac{8}{9}$ (C) 1 (D) $\frac{9}{8}$ (E) $\frac{4}{3}$

7. Edwin is standing at the point $(1, 1)$ on the coordinate plane. He shoots a laser at a mirror marked by the line $y = 4$. If he wishes for the laser to strike a sensor located at $(13, -5)$, he must aim at the point $(x, 4)$ on the wall. What is the value of x ?

(A) $\frac{7}{2}$ (B) 4 (C) $\frac{9}{2}$ (D) 5 (E) $\frac{11}{2}$

8. Let w and z be two complex numbers satisfying $w^2 + z^2 = 0$ and $w + z = 24 + 8i$. What is the value of $\operatorname{Re}(w) \cdot \operatorname{Re}(z)$?

(A) 32 (B) 64 (C) 96 (D) 128 (E) 160

9. Which of the following is closest to the area of the region bounded by $x^{1000} + y^{1000} = 1$ on the Cartesian plane?

(A) $\frac{\pi}{2}$ (B) $2\sqrt{2}$ (C) π (D) 4 (E) $3\sqrt{2}$

10. Andrew, Bob, and Charlie decide to split 6 pennies, 8 dimes and 9 quarters (not necessarily fairly). How many ways are there to distribute the coins so that each person receives at least as many quarters as dimes and at least as many dimes as pennies?

(A) 324 (B) 360 (C) 480 (D) 504 (E) 625

11. Let a, b , and c be the roots to the polynomial $P(x) = x^3 - 3x + 1$. Which of the following gives the monic polynomial with roots a^2, b^2 , and c^2 ?

(A) $P(x^2)$ (B) $P(x^2 - x)$ (C) $P(x - 2)$ (D) $P(x - 1)$ (E) $P(x + 1)$

12. Suppose the two circles $(x - 1)^2 + (y - 2)^2 = 3^2$ and $(x - 2)^2 + (y - 3)^2 = 4^2$ intersect at two points, P and Q . What is the y -intercept of the line through P and Q ?

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{2}}{2}$ (D) 1 (E) $\sqrt{2}$

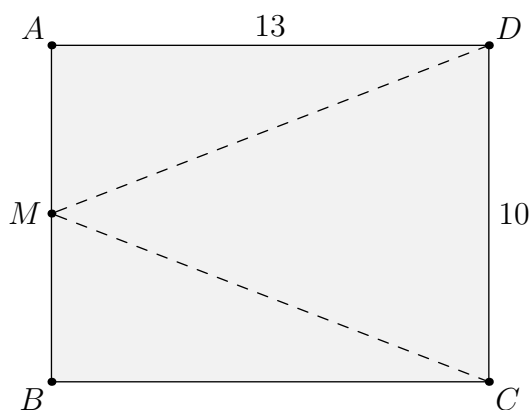
13. How many ordered pairs of positive integers (a, b) satisfy $a^2 + b = a(2019 - b)$?

- (A) 4 (B) 8 (C) 10 (D) 11 (E) 12

14. Doble writes two 1s on a whiteboard. Every second, Doble computes the sum of each pair of adjacent numbers, writing the result in between them. For example, after one second the whiteboard reads 121, and after two seconds the whiteboard reads 13231. Which of the following is closest to the sum of the numbers on the whiteboard after ten seconds?

- (A) 50,000 (B) 60,000 (C) 70,000 (D) 80,000 (E) 90,000

15. Rectangle $ABCD$ is a negligibly thin piece of paper with $AB = 10$ and $BC = 13$ sitting atop the surface of a table. Let M be the midpoint of $\overline{AB}$. Creases are made through $\overline{CM}$ and $\overline{DM}$, forming two triangular flaps. These two flaps are folded upwards so that $\overline{AM}$ and $\overline{BM}$ coincide above the surface of the table. What is the height of the resulting structure?



- (A) 4 (B) $17/4$ (C) $24/5$ (D) $60/13$ (E) $65/12$

16. How many ways are there to shade a total of 8 unit squares in a 4×4 grid such that there are exactly 2 shaded squares in each row and each column?

(A) 72 (B) 84 (C) 90 (D) 96 (E) 108

17. Let N be the number of 5-digit multiples of 24 that do not contain the digit 0. How many positive divisors does N have?

(A) 8 (B) 12 (C) 18 (D) 24 (E) 36

18. A man is standing at $(0,0)$ in a desert and needs water. There are 4 oases nearby, shaped as squares with side length s , centered at $(1,1)$, $(-1,1)$, $(-1,-1)$, and $(1,-1)$, whose sides are parallel to the coordinate axes. The man doesn't know where the oases are, so he picks a direction at random and walks forever. The probability that he eventually encounters an oasis is $1/3$. What is the value of s ?

(A) $\frac{\sqrt{3}}{8}$ (B) $2 - \sqrt{3}$ (C) $4 - 2\sqrt{3}$ (D) $\sqrt{3} - 1$ (E) $\frac{\sqrt{3}}{2}$

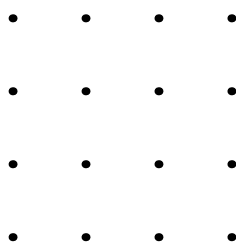
19. Let $f(a,b)$ denote the smallest positive integer c so that $\gcd(a,c)$ and $\gcd(b,c)$ are both greater than 1. Suppose x,y,z are positive integers such that $f(x,y) = 55$ and $f(y,z) = 77$. What is the number of possible values of $f(x,z)$?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

20. In $\triangle ABC$, $AB = 5$, $BC = 4$, and $AC = 6$. A circle passing through A and B intersects BC and AC at P_1 and P_2 , respectively, such that the value of $AP_1 + BP_2$ is as small as possible. The value of $AP_2 + BP_1$ can be written as m/n , where m and n are relatively prime integers. What is $m + n$?

(A) 29 (B) 31 (C) 39 (D) 41 (E) 43

21. In the array of points below, each point is one unit away from its neighbours. How many ways are there to select 4 of these points such that the distance between any two chosen points is at most $2\sqrt{2}$?



(A) 224 (B) 445 (C) 504 (D) 560 (E) 841

22. The sequence $F_1, F_2, F_3, \dots$ is defined by $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for all $n \geq 2$. Consider the quadratic polynomial

$$P(x) = x^2 + F_{100}x - F_{100}^2.$$

If the positive difference of the solutions to $P(x) = 1$ is N , what is the remainder when N is divided by 11?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 10

23. Let $p(x)$ be a polynomial with degree 6 such that $p(x) \geq 2017$ for all real numbers x . Suppose that $p(2014) = p(2015) = p(2016) = 2017$ and $p(2017) = 2018$. What is the value of $p(2019)$?

(A) 2018 (B) 2019 (C) 2020 (D) 2033 (E) 2117

24. Let n be a positive integer. On a circle there are n points, each colored black or white. Let a be the number of pairs of adjacent white dots and b be the number of pairs of adjacent black dots. Suppose there are at least 2019 distinct ordered pairs (a, b) achieved over all such colorings. What is the least possible value of n ?

(A) 45 (B) 63 (C) 64 (D) 65 (E) 90

25. In rectangle $ABCD$, $AB = 35$ and $BC = 56$. Points E and F lie on $\overline{AB}$ and $\overline{AD}$ respectively such that $5BE = 8AF$, and P is the intersection of $\overline{CE}$ and $\overline{BF}$. When the length AP attains its minimum value, AF can be expressed in the form $a\sqrt{b} - c$, for positive integers a, b, c , where b is not divisible by the square of any prime. What is $a + b + c$?

(A) 42 (B) 45 (C) 49 (D) 59 (E) 76

Time limit: 75 minutes